

Schema — Chat With Your NBA Data

Dataset

Source: [wyattowalsh/basketball](https://www.kaggle.com/wyattowalsh/basketball) on Kaggle — a pre-built SQLite database of NBA game, team, player, and draft data.

The full database has 16 tables. This project uses a trimmed subset of 6, chosen to cover team performance, game results, attendance, and draft/player bio questions while keeping the schema small enough for reliable NL-to-SQL generation.

This same information is also available live in the app itself — click the "Schema" button in frontend.html to see every table, its columns, and its current row count, without needing to read this file.

Tables in use

Team

One row per franchise.

id, full_name, abbreviation, nickname, city, state, year_founded

Game

One row per game. Team-level box score stats for both home and away sides.

game_id, season_id, season_type, game_date,

team_id_home, team_abbreviation_home, team_name_home,

wl_home, pts_home, reb_home, ast_home, fg3m_home,

team_id_away, team_abbreviation_away, team_name_away,

wl_away, pts_away, reb_away, ast_away, fg3m_away

wl_home/wl_away are 'W' or 'L'. season_type distinguishes 'Regular Season', 'Playoffs', 'Pre Season', and All-Star games (see known issues below).

game_info

Attendance data, joins to game on game_id.

game_id, game_date, attendance

common_player_info

Player bio and career info. **No scoring stats.**

person_id, first_name, last_name, position,

team_id, team_abbreviation, draft_year, season_exp,

from_year, to_year

draft_history

One row per drafted player.

person_id, player_name, season, round_number, round_pick,

overall_pick, team_abbreviation

player

Basic player lookup. **No scoring stats.**

id, full_name, is_active

Known limitation: no player scoring stats

There is no player-season or player-career stats table in this dataset. Points/rebounds/assists per player only exist buried inside play_by_play(13.6M rows of play-by-play text), which is excluded from this project's schema as impractical to aggregate reliably.

As a result, questions like "who scored the most points in a career" are **correctly refused** by the app rather than answered with invented data — see the demo question list below.

Known data quality issues (found via testing)

Two of the three issues below are now fixed by running clean_data.py(pandas) once against a fresh copy of nba.sqlite, before starting the app. The script is idempotent — running it twice doesn't cause problems, it just finds nothing left to fix the second time.

1. **Attendance outlier — FIXED by** clean_data.py: game_info.attendance contained at least one clearly erroneous value — 200,049 for game_id 0029400853, which is not physically possible for any NBA arena. clean_data.py nulls out any attendance value

over 25,000 (a generous upper bound; the largest NBA arenas seat ~20,000-21,000) rather than guessing a "correct" replacement.

2. **Inconsistent All-Star labeling — FIXED by** `clean_data.py`: `season_type` contained both 'All Star' and 'All-Star' as separate string values for the same category, which split what should be one group into two under GROUP BY `season_type`. The script normalizes both to 'All-Star'.
3. **All-Star/exhibition games mixed into "all-time" superlatives — NOT fixed, documented only**: Because game contains All-Star Game rows alongside real franchise games, "all-time" queries about single-game records (most points, most three-pointers, most wins) can surface All-Star team results (e.g. "Team LeBron") rather than actual franchises, unless the query explicitly filters `season_type` IN ('Regular Season', 'Playoffs'). This showed up across three separate demo questions during testing. Fixing this properly would mean deciding whether All-Star games belong in the queryable schema at all, which felt like a bigger call than the other two fixes — so I left it as a documented limitation instead of silently dropping data.

Demo questions

1. Which team had the best regular-season record in the 2015-16 season?
2. Which team scored the most total points in a single game, all-time?
3. Compare the average rebounds per game for two named teams in a given season.
4. Which team had the highest attendance for a single game?
5. How does average attendance compare between regular season and playoff games?
6. Which team made the most three-pointers as a team in a single game?
7. What was a specific team's playoff record in a given season?
8. Who scored the most points in a career, all-time? (*deliberately unanswerable — tests the hallucination guardrail*)
9. Which draft class had the most first-round picks who played 10+ years?
10. Which team has the most total wins across all seasons in the dataset?